# The major, course by course

B.S. COMPUTER SCIENCE · SCU BULLETIN 2026–27

- is a prerequisite for
- feeds a course offered only in alternate years
- offered in alternate years (department schedule, May 2026)
- computer science course

*Hover a course to light up what it needs and what it unlocks; hover an emphasis to see its courses. CSEN and ECEN prerequisites are not drawn.*

## LOWER DIVISION · *all required*

MATH 11
**Calculus I**
Prereq: MATH 9 / CRE ≥ 76 / AP

MATH 12
**Calculus II**
Prereq: 11

MATH 13
**Calculus III**
Prereq: 12

MATH 14
**Calculus IV**
Prereq: 13

MATH 51
**Discrete Mathematics**
no prerequisite · also CSEN 19

MATH 53
**Linear Algebra**
Prereq: 13

CSCI 10
**Intro to Computer Science**
Python · weekly lab

CSCI 60
**C++ & Object-Oriented Programming**
Prereq: CSCI 10

CSCI 61
**Data Structures**
Prereq: CSCI 60

CSCI 62
**Advanced Programming**
Prereq: CSCI 60, 61

PHYS 31 · CHEM 11 · ENVS 21 · 23
**One science course**
any one of the four

CSEN 20 · 20L · ECEN 21 · 21L
**Embedded systems · logic design**
School of Engineering

## REQUIRED UPPER DIVISION

MATH 122
**Probability & Statistics I**
Pre/co-req: 14

CSCI 161
**Automata & Languages**
Prereq: MATH 51

CSCI 163
**Theory of Algorithms**
Prereq: MATH 51, CSCI 61 · also CSEN 179

CSEN 177 · 177L
**Operating Systems**
School of Engineering

## ALGORITHMS & COMPLEXITY

CSCI 162
**Computational Complexity**
Prereq: MATH 51

CSCI 164
**Advanced Theory of Algorithms**
Prereq: CSCI 163

CSCI 165
**Data Privacy & Algorithmic Fairness**
Prereq: MATH 51, CSCI 10 · by request

CSCI 146
**Optimization I**
Prereq: CSCI 10, MATH 14, 53 · also MATH 146

CSCI 147
**Optimization II**
Prereq: MATH 122, CSCI 146 · also MATH 147

## DATA SCIENCE

CSCI 183
**Data Science & Machine Learning**
Prereq: CSCI 62, MATH 53, 122

CSCI 184
**Advanced Machine Learning**
Prereq: CSCI 183

CSCI 185
**Data Mining**
Prereq: CSCI 62, MATH 53, 122

CSCI 127
**Computer Simulation**
Prereq: CSCI 10 (MATH 122 recommended) · by request

CSCI 166
**Numerical Analysis**
Prereq: CSCI 10, MATH 53 · also MATH 166

## MATH ELECTIVES · *named by the emphases*

MATH 101
**Survey of Geometry**
Prereq: 13 · algorithms emphasis

MATH 175
**Theory of Numbers**
Prereq: MATH 52 (not in the CS major) · algorithms, security

MATH 176
**Combinatorics**
Prereq: 51 · algorithms emphasis

MATH 177
**Graph Theory**
Prereq: 51 · algorithms emphasis

MATH 123
**Probability & Statistics II**
Prereq: 122; 53 pre/co · data science emphasis

## MINOR IN COMPUTER SCIENCE

MINOR
**Computer Science**
CSCI 10, 60, 61, 62; MATH 51; CSEN 20 & 20L; two upper-division CSCI and one CSCI or CSEN

## SOFTWARE

CSCI 169
**Programming Languages**
Prereq: CSCI 62, MATH 51

CSCI 187
**Design & Management of Software**
Prereq: CSCI 62

CSCI 168
**Computer Graphics**
Prereq: CSCI 62, MATH 13

## SECURITY

MATH 178
**Cryptography**
Prereq: MATH 51

CSCI 180
**Computer Security**
Prereq: CSEN 20, CSCI 62

CSCI 181
**Applied Cryptography**
Prereq: MATH 178, CSCI 62

## CSEN ELECTIVES

**146, 152, 161, 166, 169, 178**
School of Engineering courses named by the emphases; prerequisites not drawn

## EMPHASES · *choose one · five upper-division courses*

### Algorithms & Complexity
CSCI 162, 164 · two of CSCI 146, 147, 165, 181, MATH 101, 175, 176, 177, 178 · one more from that list or any upper-division CSCI/CSEN

### Data Science
CSCI 183, 184, 185 · two of CSCI 127, 146, 147, 164, 166, MATH 123, CSEN 166, or any upper-division CSCI/CSEN

### Security
MATH 178, CSCI 180, 181 · two of MATH 175, CSEN 152, 161, 146, or any upper-division CSCI/CSEN (not CSEN 150)

### Software
CSCI 169, 187, CSEN 146 · one of CSCI 183, 180, 168 or any upper-division CSCI · one of CSCI 183, 180, 168, CSEN 161, 178 or any upper-division CSCI/CSEN

### For every emphasis
Five 4- or 5-unit upper-division courses. Decide by the end of junior year; only one emphasis. Data Science cannot be the emphasis for both a math and a CS major.